



Tran Thanh Quan

Ho Chi Minh City, Vietnam | 0974426058

tranthanhquan09@gmail.com | [Facebook](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Backend-focused Software Engineering student with strong proficiency in OOP, Java, and the Spring ecosystem. Experienced in architecting scalable systems in microservices and AWS cloud environments. Adept at building CI/CD pipelines, optimizing databases, and leveraging **AI coding assistants** (Claude, Copilot) to accelerate delivery. Familiar with crafting contextual prompts and maintaining responsible AI use. Seeking the Java Software Engineer role to help KMS build brilliant, high-impact solutions.

EDUCATION

FPT University

Bachelor of Engineering in Software Engineering – GPA: 8.0/10

Ho Chi Minh City, Vietnam

Aug. 2023 – Aug. 2027

PROJECTS

Flash Sale System & Reliability Lab [GitHub](#) | *Java 21, Spring Boot, Redis, Kafka, AWS* **Dec. 2025 – Present**

- Engineered an atomic **Redis Lua** pre-deduction gate and **Spring Boot** compensation workflow, successfully processing **5,000 concurrent requests** across 100 threads with zero oversell.
- Architected a **Transactional Outbox pattern** to asynchronously stream order events to **Kafka**, guaranteeing at-least-once delivery and preventing message loss during high-load traffic spikes.
- Integrated **Spring Security** and managed automated deployments via **CI/CD pipelines** onto AWS infrastructure, utilizing GitHub Copilot and ChatGPT to quickly generate boilerplate code.

AI Hospital Knowledge Assistant [GitHub](#) | *FastAPI, PostgreSQL, pgvector, Redis, RAG* **Feb. 2026 – Present**

- Developed a **HIPAA-compliant** medical RAG chatbot using modern LLMs. Authored strict **contextual prompts** and 'fail-closed' guardrails to reject 100% of ungrounded responses for clinical safety.
- Engineered an asynchronous document ingestion pipeline to process OCR and **pgvector** embeddings, implementing robust RBAC to prevent cross-patient data leakage.
- Architected a SQL-backed Graph RAG engine in **PostgreSQL** for multi-hop entity traversal, deploying the application via CI/CD pipelines to ensure continuous integration.

Aivora – Freelance Marketplace [Link](#) | [GitHub](#) | *.NET, ReactJS, PostgreSQL, SignalR* **May. 2026 – Present**

- Spearheaded backend development in an agile team to architect a full-stack platform, integrating **ReactJS** frontend via REST APIs.
- Engineered an **idempotent** escrow payment engine with pessimistic locking, eliminating double-spending risks and ensuring 100% financial accuracy across milestone transactions.
- Built an **AI Job Assistant** to automatically evaluate fit and optimize job descriptions, accelerating client-expert negotiation cycles by 30% using real-time messaging via **SignalR**.

ACTIVITIES & CERTIFICATIONS

Tet Luon Vui – Community Fundraising Project [Facebook](#) | *Communications & Logistics* [Evidence](#) | 9 Members | 2026

- Managed campaigns and logistics, growing social audience to 900+ and exceeding fundraising goals.

Certifications & Training: [Architecting Solutions on AWS \(2026\)](#); [Retrieval Augmented Generation – RAG \(2026\)](#); Global Consumer Intelligence Data Science Program (The University of Tokyo, 2026) (In progress)

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, Python, C#, SQL, HTML/CSS

Java Ecosystem: Spring Boot, Spring Core, MVC, Security, Data, JPA, Hibernate

Frameworks & Web: ReactJS, Next.js, REST APIs, FastAPI, .NET

Cloud & DevOps: AWS (EC2, S3), Docker, CI/CD, Git, GitHub Actions, Linux

Databases & Messaging: PostgreSQL, MySQL, Redis, Kafka, pgvector

AI Tools: GitHub Copilot, Claude Code, Cursor, ChatGPT, Gemini, Prompt Engineering, RAG

Architecture: Object-Oriented Programming (OOP), Microservices, Domain-Driven Design (DDD)